

Blockchains & Distributed Ledgers

Lecture 10

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Finance

- {creation, management, investment} of **money** and **financial assets**
- Financial assets: non-physical assets whose value is derived by contractual claim
 - Bank deposits, stocks, bonds, loans
- Financial services
 - Lending/borrowing, issuing securities, managing funds
- Financial markets: marketplaces for **trading** financial assets

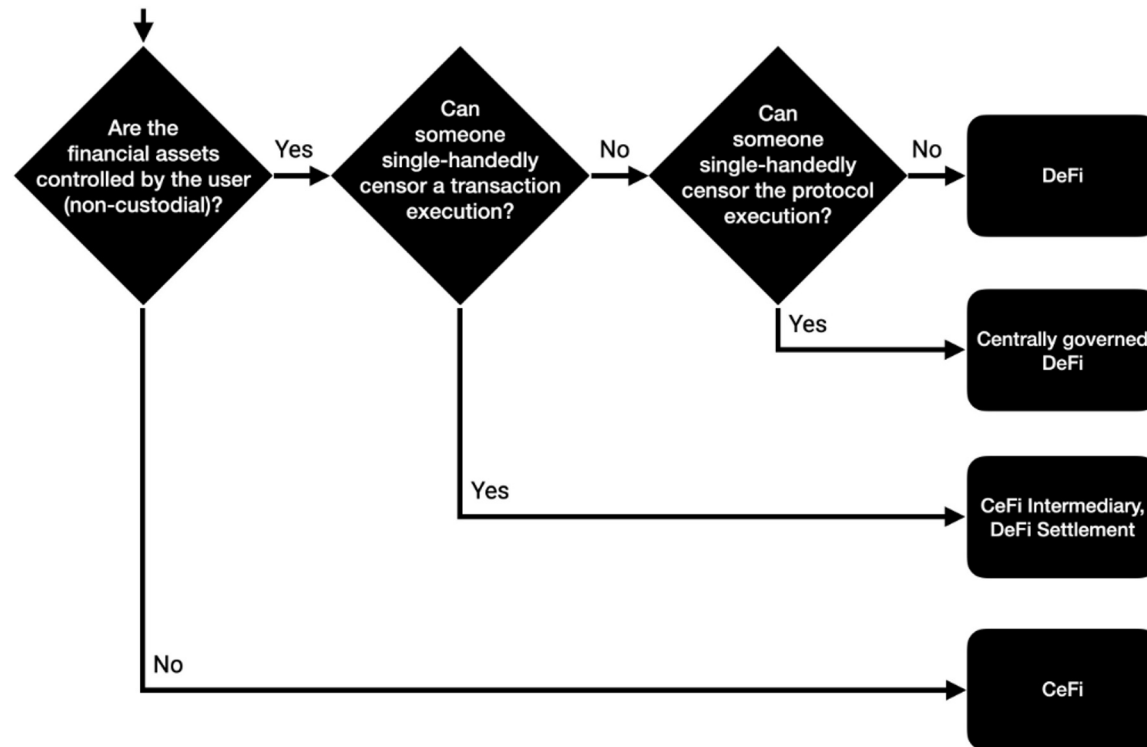
Decentralized Finance (DeFi)

- Financial products and services on decentralized infrastructure
- Do not rely on centralized intermediaries
 - E.g., exchanges, banks, brokers
- Utilize the security of an underlying blockchain system
- Open to hazards and attacks that stem from public/decentralized nature of blockchains

Securities

- Security: a fungible, negotiable, financial instrument that has some value
 - stock (representing ownership of company) - **equity security**
 - bond (representing a creditor relationship with company/government) - **debt security**
- In the US (cf. Securities and Exchange Commission (SEC) v. W.J. Howey Co) a security is:
 - a contract, transaction, or scheme whereby a person **invests** his money in a **common enterprise**...
 - *Horizontal commonality*: Investors' assets are joined and they share the risk and benefits
 - *Vertical commonality*: Investors' fortunes are linked and dependent upon the efforts of those seeking the investment (**narrow** (investors' profits rise and fall together with promoter's) vs. **broad** (investors' profits depend on promoter's expertise and performance))
 - ... and is led to **expect** profits solely from the efforts of the promoter or a third party

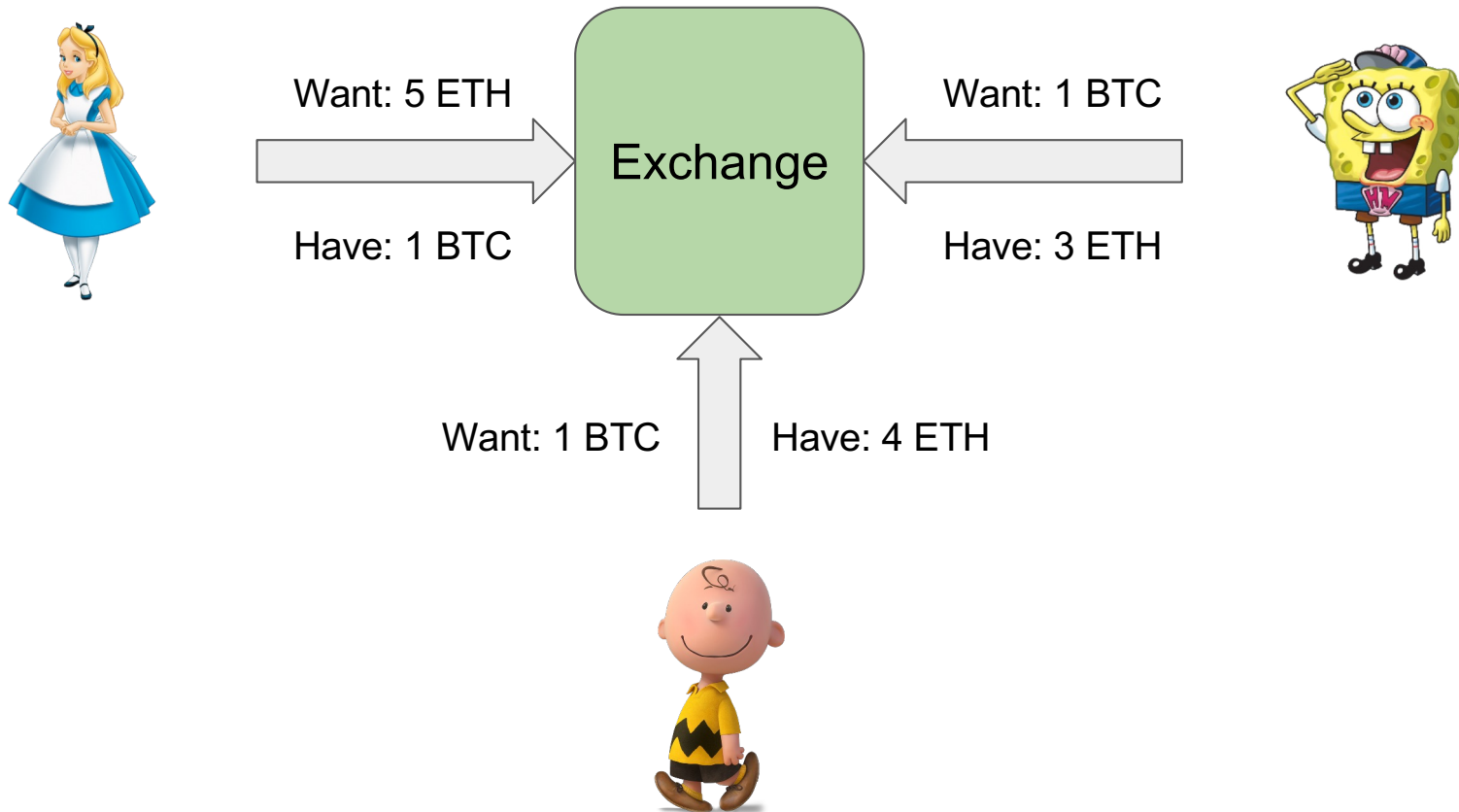
Decentralized Finance (DeFi)



Source: <https://berkeley-defi.github.io/f21>

Exchanges

Exchanges/Marketplaces



Decentralized Exchanges (DEXs)

- Completely on-chain
 - Trades between native chain currency (e.g., ETH) and on-chain tokens (e.g., ERC20)
- Censorship resistance
 - Availability depends on underlying blockchain's safety & liveness
- Differences from centralized (server-based) exchanges
 - (Blockchain) fees for creating orders
 - (Blockchain) fees for cancelled orders
 - Slower matching
 - No KYC and AML provisions

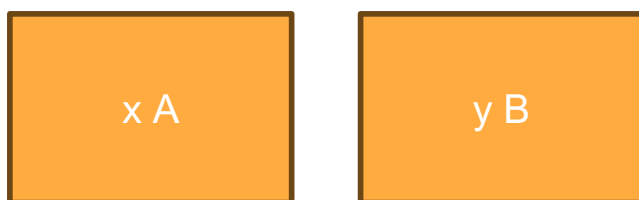
Order book based Exchange

Pair X-Y



Automated Market Makers

A-B pair

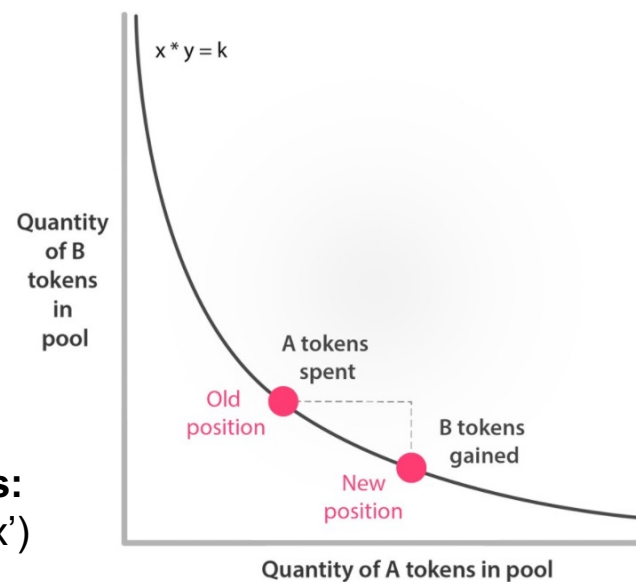


Invariant $x * y = k$

Convert x' amount of A tokens into B tokens:
 $\Rightarrow$ New state $(x+x')*(y-y') = k \Leftrightarrow y' = x'y/(x + x')$

If $x' \ll x$ then $y' \approx x' (y/x)$
(but can be far from y/x otherwise)

“Liquidity providers” are incentivized to keep x, y big



Decentralized Exchanges (DEXs), Attacks

- Front-running
 - Adversary can use gas price to front-run a trading tx
 - Miners choose tx ordering → can front-run plain users
 - Also exists in centralized exchanges (esp. if unregulated)
 - Exchange owner can see all txs, control execution order, and increase/decrease price arbitrarily to “burn” customers (both short and long)
- Some mining pools offer front-running *as a feature* (e.g., [Ethermine](#))

Decentralized Exchanges (DEXs), Attacks

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- Insertion (aka sandwich) attack
 - U creates a “buy” order TX_U , e.g., buy BTC for ETH (expectation: BTC-ETH will go up)
 - Attacker inserts before TX_U (front-running) another buy order and gets x BTC for y_1 ETH, *moving the price* of BTC-ETH up
 - U’s order is executed for the increased price
 - Attacker inserts a sell order for BTC-ETH after TX_U , which gets back y_2 ETH for x BTC
 - Attack profit: $y_2 - y_1 > 0$

Loans

(Real-world) Loans



Borrower

Request loan for \$x



Lender
(Bank)

(Real-world) Loans



Borrower

Request loan for \$x



Check, estimate default risk, (perhaps) require collateral



Lender
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(Real-world) Loans



Borrower

Request loan for $\$x$

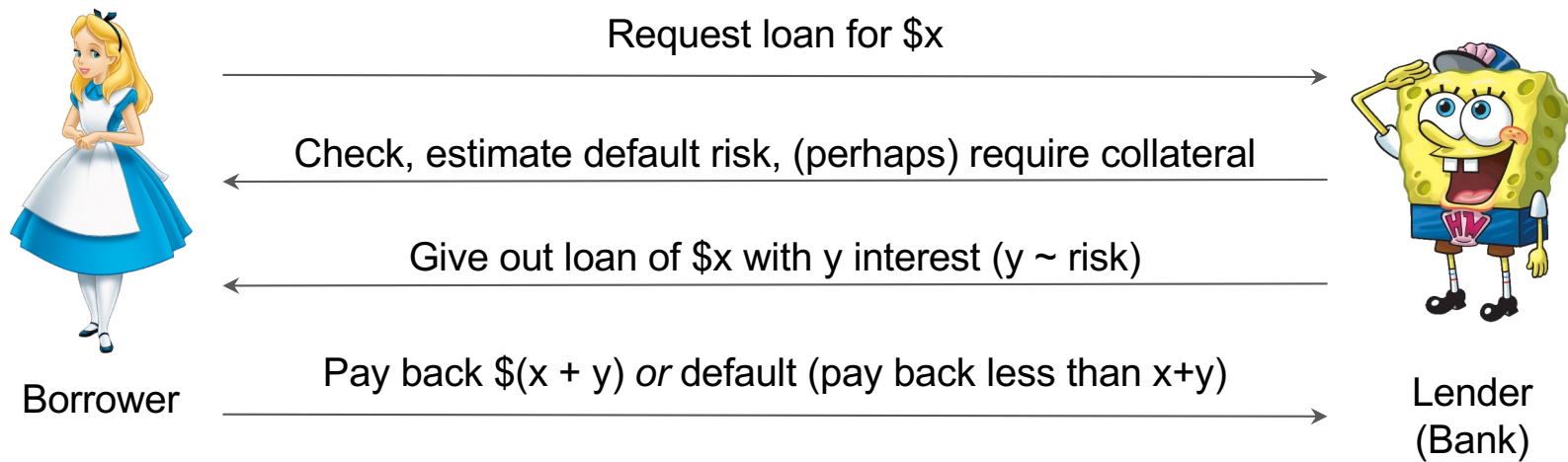
Check, estimate default risk, (perhaps) require collateral

Give out loan of $\$x$ with y interest ($y \sim \text{risk}$)



Lender
(Bank)

(Real-world) Loans



Decentralized Loans

- Assumption: **Oracle** that reports (real-world) asset prices
 - E.g., USD prices
 - Typically semi or completely centralized
- Lender deposits principal **capital** to a “vault”
 - The vault is simply the service’s smart contract
- Borrower deposits **collateral** to borrow from vault
 - Typically over-collateralized: $\text{value}(\text{collateral}) \text{ (in real prices) } > \text{value}(\text{loan})$
 - If collateral value drops significantly, loan can be automatically liquidated
 - Liquidator repays debt and gets the collateral at a discount
- Borrower returns loan + interest to vault
 - Lender can withdraw principal capital and received interest

Flash Loans

- A loan that occurs in a **single atomic** transaction
- Lender adds principal capital (“liquidity”) to a smart contract pool
- Within a single transaction:
 - Smart contract pool transfers x assets from the pool to borrower’s account
 - Borrower uses x assets as they want
 - Borrower transfers x assets plus some fee to the pool
 - If any step of the above fails (e.g., borrower cannot repay the pool), tx fails
- No default risk!

Decentralized/Flash Loans, Attacks

- Price oracle manipulation
 - Control collateral requirements

Decentralized/Flash Loans, Attacks

- Price oracle manipulation
 - Control collateral requirements
- Risk-free arbitrage
 - DEXs may offer different prices on the same trading pair
 - Use flash loan to:
 - i) buy on one DEX
 - ii) sell on the other (at higher price)
 - iii) repay loan+fees and profit depending on price difference

Decentralized/Flash Loans, Attacks

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- Washtrading
 - Sell and buy the same asset to create misleading activity, e.g., to artificially increase trading volume and show “demand”
 - Centralized cryptocurrency exchanges often perform washtrading (e.g., [1](#), [2](#), [3](#))
 - Illegal in USA *regulated* markets since 1936

Market Capitalization

Market capitalization (of cryptocurrencies)

- Centralized exchanges are sources of price
 - Price of X: the latest price for which a single X token was sold (in exchange for USD/GBP/Bitcoin/altcoins/...)
- Market cap: $\text{<number of coins in circulation>} \cdot \text{<price>}$



<u>Product name</u>	<u>Circulating Tokens</u>	<u>Price</u>	<u>Market Cap</u>	<u>Total MC</u>



1 BTC



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Bitcoin	1			



1 BTC

Sell 1 BTC for \$1



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Bitcoin	1	\$1 (1BTC=1\$)	\$1	\$1



\$1



1 BTC



1 ETH

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Bitcoin	1	\$1 (1BTC=1\$)	\$1	\$1
Ethereum	1			



\$1



1 ETH

Sell 1 ETH
for 1 BTC



1 BTC

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\$1



1 ETH



1 BTC

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Bitcoin	1	\$1 (1BTC=1\$)	\$1	\$2
Ethereum	1	\$1 (1 ETH=1BTC)	\$1	



\$1



1 BTC

Buy 0.5 ETH
for 1 BTC



1 ETH

<u>Product name</u>	<u>Circulating Tokens</u>	<u>Price</u>	<u>Market Cap</u>	<u>Total MC</u>
Bitcoin	1	\$1 (1BTC=1\$)	\$1	\$2
Ethereum	1	\$1 (1 ETH=1BTC)	\$1	



\$1



0.5 ETH,
1 BTC



0.5 ETH

<u>Product name</u>	<u>Circulating Tokens</u>	<u>Price</u>	<u>Market Cap</u>	<u>Total MC</u>
Bitcoin	1	\$1 (1BTC=1\$)	\$1	\$3
Ethereum	1	\$2 (1ETH=2BTC)	\$2	



\$1

Buy 0.5 BTC for \$1



0.5 ETH,
1 BTC



0.5 ETH

<u>Product name</u>	<u>Circulating Tokens</u>	<u>Price</u>	<u>Market Cap</u>	<u>Total MC</u>
Bitcoin	1	\$1 (1BTC=1\$)	\$1	\$3
Ethereum	1	\$2 (1ETH=2BTC)	\$2	



0.5 BTC



0.5 ETH,
0.5 BTC,
\$1



0.5 ETH

<u>Product name</u>	<u>Circulating Tokens</u>	<u>Price</u>	<u>Market Cap</u>	<u>Total MC</u>
Bitcoin	1	\$2 (1BTC=2\$)	\$2	\$6
Ethereum	1	\$4 (1ETH=2BTC)	\$4	



0.5 BTC



0.5 ETH,
0.5 BTC,
\$1



0.5 ETH

<u>Product name</u>	<u>Circulating Tokens</u>	<u>Price</u>	<u>Market Cap</u>	<u>Total MC</u>
Bitcoin	1000	\$2 (1BTC=2\$)	\$2000	\$6000
Ethereum	1000	\$4 (1ETH=2BTC)	\$4000	

Cryptos: 21,777 Exchanges: 524 Market Cap: \$825,029,479,545

1	 Bitcoin BTC	\$16,587.72	▲0.07%	▼0.39%	▲0.49%	\$318,646,947,717	\$31,806,610,049 1,917,154 BTC	19,209,806 BTC
2	 Ethereum ETH	\$1,201.47	▲0.26%	▼1.49%	▲0.78%	\$147,028,970,200	\$11,129,826,172 9,258,015 ETH	122,373,866 ETH
3	 Tether USDT	\$0.9996	▲0.00%	▲0.03%	▲1.26%	\$65,917,967,109	\$42,097,899,159 42,115,255,827 USDT	65,944,685,876 USDT
4	 USD Coin USDC	\$1.00	▲0.00%	▲0.02%	▼0.62%	\$44,417,530,170	\$3,698,812,883 3,698,369,386 USDC	44,406,592,473 USDC
5	 BNB BNB	\$267.49	▲0.13%	▼1.44%	▼4.07%	\$42,791,727,100	\$933,939,060 3,489,744 BNB	159,973,721 BNB
6	 Binance USD BUSD	\$1.00	▼0.05%	▼0.05%	▼0.96%	\$23,039,136,412	\$6,672,905,015 6,671,289,989 BUSD	23,037,140,170 BUSD

11/2022

Exchanges: 761 Market Cap: \$3.07T ▼ 0.20% 24h Vol: \$182.35B ▼ 7.45% | <https://coinmarketcap.com>

1	 Bitcoin BTC	\$91,819.95	▼ 0.73%	▲ 1.03%	▲ 3.79%	\$1,816,630,201,968	\$74,822,656,662 814,414 BTC	19,784,700 BTC
2	 Ethereum ETH	\$3,107.98	▲ 0.31%	▼ 1.23%	▼ 5.26%	\$374,279,710,187	\$30,650,350,735 9,872,454 ETH	120,425,284 ETH
3	 Tether USDT	\$1.00	▼ 0.00%	▲ 0.05%	▲ 0.06%	\$128,309,736,183	\$146,176,620,783 145,999,506,644 USDT	128,151,427,237 USDT
4	 Solana SOL	\$236.00	▼ 0.31%	▼ 1.39%	▲ 10.02%	\$112,021,424,893	\$6,695,695,332 28,445,626 SOL	474,665,742 SOL
5	 BNB BNB	\$614.61	▲ 0.60%	▼ 0.56%	▼ 2.80%	\$88,510,147,266	\$2,060,297,347 3,358,727 BNB	144,010,770 BNB
6	 XRP XRP	\$1.10	▲ 1.61%	▼ 1.94%	▲ 53.04%	\$62,796,841,029	\$5,882,343,762 5,361,958,677 XRP	56,931,242,174 XRP
7	 Dogecoin DOGE	\$0.3875	▼ 0.48%	▲ 3.73%	▼ 0.87%	\$56,910,962,036	\$12,796,614,942 33,087,624,904 DOGE	146,861,506,384 DOGE
8	 USDC USDC	\$1.00	▲ 0.00%	▲ 0.01%	▲ 0.03%	\$37,366,228,469	\$10,928,604,894 10,929,574,772 USDC	37,357,540,602 USDC
9	 Cardano ADA	\$0.7351	▼ 0.21%	▼ 1.43%	▲ 25.87%	\$25,754,295,388	\$1,558,327,158 2,118,515,142 ADA	35,033,776,787 ADA
10	 TRON TRX	\$0.1993	▼ 0.27%	▼ 1.38%	▲ 6.70%	\$17,212,615,787	\$817,697,111 4,104,191,687 TRX	86,364,116,154 TRX

20/11/2024

Market capitalization (of cryptocurrencies)

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 - Price of X: the latest price for which a single X token was sold (in exchange for USD/GBP/Bitcoin/altcoins/...)
- Market cap: $\text{<number of coins in circulation>} \times \text{<price>}$

Issues

- Market cap may be artificially increased
 - E.g., tokens or dubious “coins” sold for other cryptocurrency
- Question: What is the ratio of *real-world* money to *market cap*? In other words, how much *real-world* money is *actually* in the market?

Stablecoins

Fiat-backed stablecoins

- Centralized issuer of “stable price” tokens
- How it works
 - User deposits \$1 to service’s bank account
 - Service issues 1 token in exchange
 - *As long as* token remains in circulation, the service keeps \$1 in escrow
 - *Whenever* user wants, they can redeem 1 token for \$1
- Why use such stablecoins instead of USD directly?
 - Exchanges
 - Simplify regulation compliance
 - settle inter-exchange transfers fast
 - Users
 - Engage in DeFi
 - bypass capital controls
 - avoid KYC/AML requirements

Fiat-backed stablecoins

- If 1-1 promise (silently) breaks
 - Service issues loans (fractional reserve), assuming default risk
 - Service can insert (artificial) liquidity into the market (to pump price/market cap of assets)
- If regulation tightens
 - The broken 1-1 promise becomes public knowledge
 - Trust in the system decreases, “stable” price no longer stable (reflecting default risk)
 - Liquidity evaporates due to “firesale” of stablecoin
- Tether (*by far* the largest “stablecoin”)
 - Opaque (*no audits*, unknown reserves, unknown affiliations, can refuse redemptions at will)
 - misleading behaviour ([NYAG](#), [CFTC](#))
 - *It is known* that Tether does not have \$1 for every USDT
 - Circulation: \$4B till 2019, \$21B 12/2020, \$74B in 11/2021, \$65.9B in 11/2022, \$132B 11/2024
 - “Daily trading volume” across all exchanges: \$171B (>2x Bitcoin’s)
 - Almost every major exchange trades Tether (and is open to Tether collapse risk)

Crypto-backed stablecoins

- $(1+x)$ -1 backing by crypto reserves
- (Centralized) price oracles
- How it works
 - Assume: 1 ETH = \$1, $x = 1$
 - Deposit 2 ETH and mint 1 stablecoin (over-collateralized)
 - If $\text{price(ETH)} > \$0.5$: stablecoin's price unchanged
 - If $\text{price(ETH)} < \$0.5$: stablecoin liquidated, investor receives 2 ETH
- Example: Dai

Crypto-backed stablecoins

- Leveraged investment
 - a. Mint 1 coin with 2 ETH
 - b. Buy 1 ETH with 1 coin (price of ETH is \$1)
 - c. Increased demand for ETH → ETH price ↑
 - d. ETH price ↑ (eg. 1 ETH = \$2) → sell 0.5 ETH for 1 coin, redeem coin for 2 ETH (profit: 0.5 ETH)
 - e. Go to (a) (price pump)
- What if ETH price drops?
 - a. Stablecoins liquidated for ETH
 - b. Investors sell ETH to cut losses → Uncertainty from liquidations, ETH supply ↑ → price ↓
 - c. Go to (a) (death spiral)
- Example: March 2020, MakerDAO had to *centrally intervene and inject liquidity* to avoid complete shutdown
 - What happens if market collapses and external pockets not deep enough?

Algorithmic stablecoins

- (Premise) a price oracle
- Two types of assets
 - “stable” coins
 - bonds
- How it works
 - coin price > \$1: automatically issue and distribute new coins (coin supply $\uparrow \rightarrow$ price $\downarrow$)
 - coin price < \$1: sell bonds for coins (coin supply $\downarrow \rightarrow$ price $\uparrow$)
- Bonds:
 - Buy bond in auction (face value: \$1, auction price: y)
 - If coin price gets above \$1 again, redeem bond to receive new coins (profit = $1 - y$)

* Irving Fisher. “The Purchasing Power of Money” (1911)

Algorithmic stablecoins

- No such project has survived for long
 - Nubits (*“World's Best Stable Digital Currencies”*): \$0.12
 - Basis (*“an Algorithmic Stablecoin Pegged to 1 USD”*): \$0.04
 - Terra (*“stable rewards in all economic conditions”*): \$0.02
- Why fail?
 - price $\uparrow$ $\rightarrow$ bond-holders and investors receive newly minted coins
 - price $\downarrow$ $\rightarrow$ investors can only buy bonds and *have faith* that price will go up again
 - if price does not go up quickly
 - lost profit (opportunity cost) $\uparrow$
 - if lost profit > bond profit, no reason to remain invested

Applications

Use an independent DL or piggyback on existing?

<i>Scheme</i>	Advantage	Disadvantage
<i>Piggybacking</i>	Potential for higher assurance	Need to engineer or program protocol rules into existing ledger
<i>Independent</i>	Ability to customise protocol & enforce individual properties	Might attract a small set of initial nodes and initially be less trustworthy

Comparison of different blockchains

<i>Blockchain</i>	Mode of operation	Smart Contract Capability	Account Model	Notes
Bitcoin	Proof of Work	Limited / Bitcoin script	UTXO model	High energy consumption. First bid pricing.
Ethereum	Proof of Stake	General / Ethereum VM	Account based	Minimal energy consumption. Variable fees.
Cardano	Proof of Stake	General / Plutus scripts	Extended UTXO model	Minimal energy consumption. Fixed fees.
Solana	Proof of History	EVM compatible	Account based	Modest energy consumption. Priority fees.

Layer 2 solutions

- Blockchain “layer 1”s do not scale well
 - Throughput - transaction per second (TPS)
 - Latency – fast confirmation
- Layer 2 solutions :
 - modify some of the availability / safety characteristics of layer 1 in favor of better scaling
 - Resort to a layer 1, when disputes / problems arise.
- Opportunities
 - L2 that interoperate across L1 chains
 - “hybrid Dapps” that operate with different L1’s

Digital economy (on a blockchain)

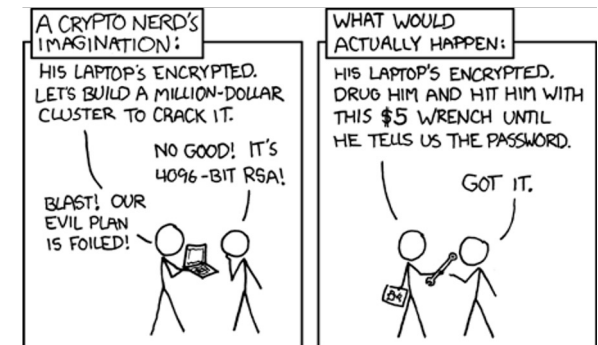
- Use a blockchain to record monetary transactions
- Create new money based on pre-determined algorithm

Digital economy (on a blockchain)

- Use a blockchain to record monetary transactions
- Create new money based on pre-determined algorithm

Issues

- Why would people use on-chain tokens as *money* instead of as commodities?
Why would someone sell BTC, if they expect its (USD) price to increase?
- How to accurately value a blockchain-based economy? (e.g., market capitalization)



Name registry (on a blockchain)

- Use a blockchain to register names
- Useful in the context of DNS (domain name system) and public-key directories
- Censorship-resistant
- Examples:
 - *Namecoin*: separate blockchain, based on Bitcoin protocol
 - *Blockstack*: piggybacking on the Bitcoin blockchain
 - *ENS (Ethereum Name Service)*: domain registry implemented as an Ethereum smart contract

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Issues

- How to connect blockchain-issued names with the rest of the internet?
- What if some domains *should be* taken down?

Land ownership (on a blockchain)

- Issue a new digital asset linked to land title
- Store information in the digital asset that links to an information resource
 - E.g., insert a URL to real-world registry or an identifier for a torrent file
- Digital asset becomes representation of ownership
 - He who controls the asset can prove or transfer ownership of the linked land
- Same idea can be extended to any real-world asset

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- Same idea can be extended to any real-world asset

Issues

- What happens if the information source is no longer available (e.g., the URL breaks)?
- What if the legal system does not recognize on-chain representation?

Local Economies (on a blockchain)

- Small communities could try to protect their sovereignty and economical independence via a decentralised local economy on a blockchain

Issues

- Small economies often struggle to sustain themselves
- Global market often outcompetes them

Supply chain tracking (on a blockchain)

- Real-world products
 - E.g., clothes, shoes, meat, olive oil, even diamonds
- Create a digital fingerprint of the object
- Register the fingerprint on a blockchain
- Record every change in the object's state
 - E.g., creation at source, transportation, selling/buying

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- Record every change in the object's state
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Issues

- How to create a fingerprint (unique digital representation) of a physical object?
- How to make sure that people that handle the object actually record its state changes properly?

Philanthropy (on a blockchain)

- An NGO/philanthropic organization creates a smart contract
 - E.g., to collect funds for building a school
- People send funds to the contract
- The contract keeps the funds in escrow:
 - When a proof that the project is complete is provided, the contract releases the funds
 - If a deadline passes, the remaining funds are returned to the participants

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Issues

- What kind of (secure) proofs of real-world actions could be understandable by a smart contract?
- How can you prevent embezzlement, i.e., a corrupted official publishing incorrect proofs?

Crowdfunding (on a blockchain)

- A project creates a smart contract that issues tokens
 - Initial Coin Offering (ICO), ERC20 Ethereum tokens
- Users give coins in exchange for tokens
 - Buy tokens with ETH
- Tokens can:
 - Be used in a future platform that the project creates (utility tokens)
 - Be used as investment (securities)

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Issues

- How to guarantee that project will not run away with the funds (i.e., exit scam)?
- What if promoters scam investors and authorities, e.g., via a pump and dump?
- Are the promises of the project verified/regulated? Will the project face penalties for lying?

Prediction Markets

- A market that enables trading on future events (e.g. insurance)
- Oracles provide real-world information on whether an event occurred
- Example: “10 tornadoes will hit USA in 2020”
 - participants bet in favour or against the event
 - market shares: YES = α , NO = $1-\alpha$; total investment: X ; probability of event happening: p
 - YES (NO) participants receive X iff event (doesn't) happen. Expected Profit of YES = $pX - \alpha X$

Prediction Markets

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Issues

- Trust in the oracle? Can a decentralized oracle for real-world information exist?

IoT and micropayments (on a blockchain)

- IoT devices connected to the internet (e.g., smart fridges, sensors)
- Utility meters (e.g., electricity or water consumption)
- User pays in real-time with multiple “micro”-payments to the service provider
- Alternative to subscription model
- Monetization of user data: User gets income for selling their personal data

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- User pays in real-time with multiple “micro”-payments to the service provider
- Alternative to subscription model
- Monetization of user data: User gets income for selling their personal data

Issues

- fees can increase dramatically near congestion
- why would you share your daily data with the whole world? Would you sell your personal life?

Gaming and art collection (on a blockchain)

- In-game currency on a blockchain
 - E.g., Ethereum-based game tokens
- Digital collectibles
 - E.g., trading cards, virtual animans (CryptoKitties), NFTs (Non-Fungible Tokens) of art works
- On-chain games
 - Gambling, strategy games, social network games, ...

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Issues

- Gaming companies typically want control of in-game economy - why would decentralization benefit them?
- Why would users pay fees to play, when centralized options are free (or, at worst, pay-to-win)?

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